

LAB EXPERIMENT RECORD

Set 1: Monthly Sales Data

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Tuesday

Aim

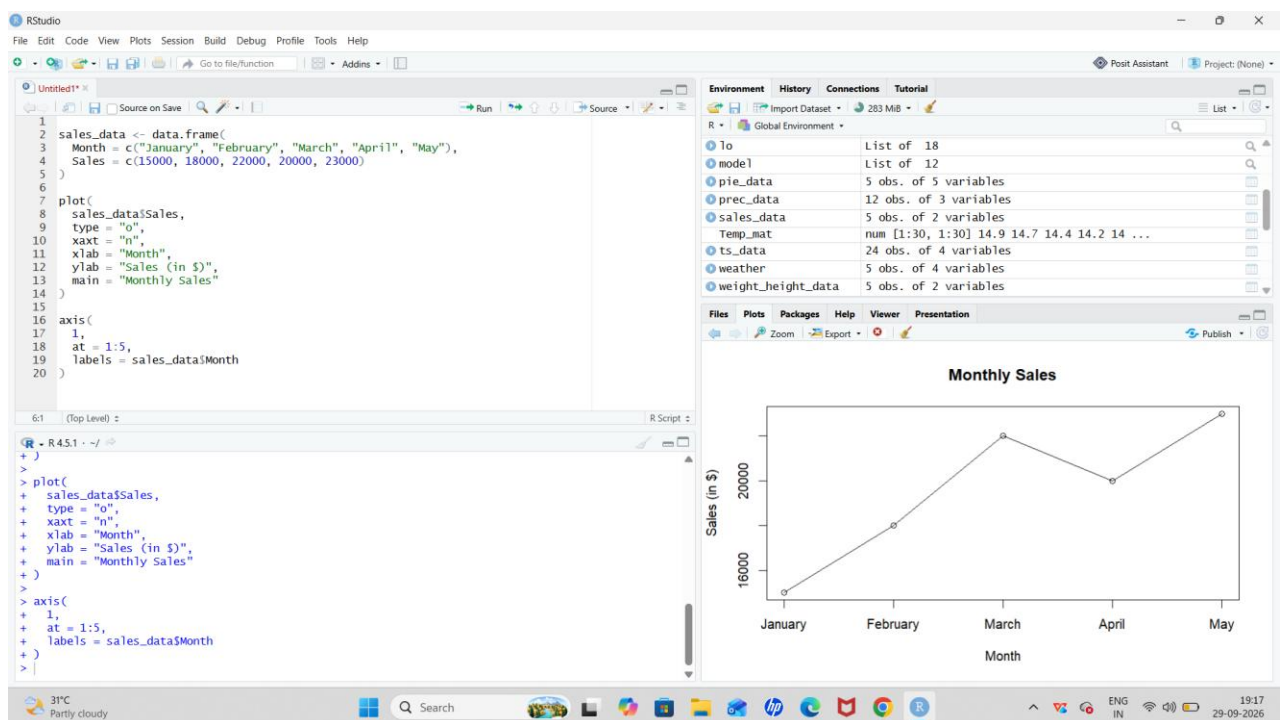
To analyze monthly sales data using **R programming** by creating a line chart, bar chart, scatter plot, and an interactive dashboard to understand sales trends, product performance, and the relationship between advertising budget and sales.

Dataset: Monthly Sales

Month	Sales (in \$)
January	15000
February	18000
March	22000
April	20000
May	23000

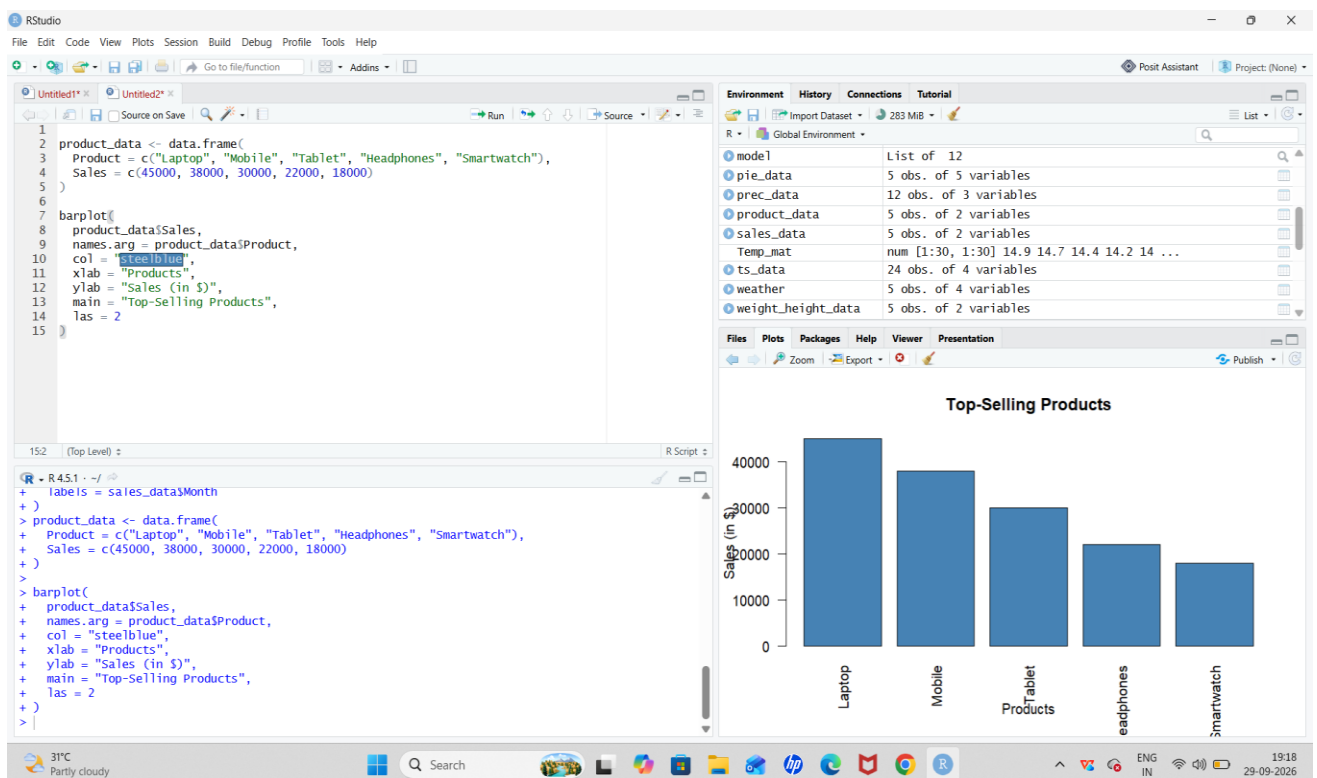
1. Create a line chart to visualize the monthly sales. Label the axes and title the chart appropriately.

```
sales_data <- data.frame(
  Month = c("January", "February", "March", "April", "May"),
  Sales = c(15000, 18000, 22000, 20000, 23000)
)
plot(
  sales_data$Sales,
  type = "o",
  xaxt = "n",
  xlab = "Month",
  ylab = "Sales (in $)",
  main = "Monthly Sales"
)
axis(
  1,
  at = 1:5,
  labels = sales_data$Month
```



2. Generate a bar chart to display the top-selling products for the year. Label the chart elements.

```
product_data <- data.frame(  
  Product = c("Laptop", "Mobile", "Tablet", "Headphones", "Smartwatch"),  
  Sales = c(45000, 38000, 30000, 22000, 18000)  
)  
barplot(  
  product_data$Sales,  
  names.arg = product_data$Product,  
  col = "steelblue",  
  xlab = "Products",  
  ylab = "Sales (in $)",  
  main = "Top-Selling Products",  
  las = 2  
)
```

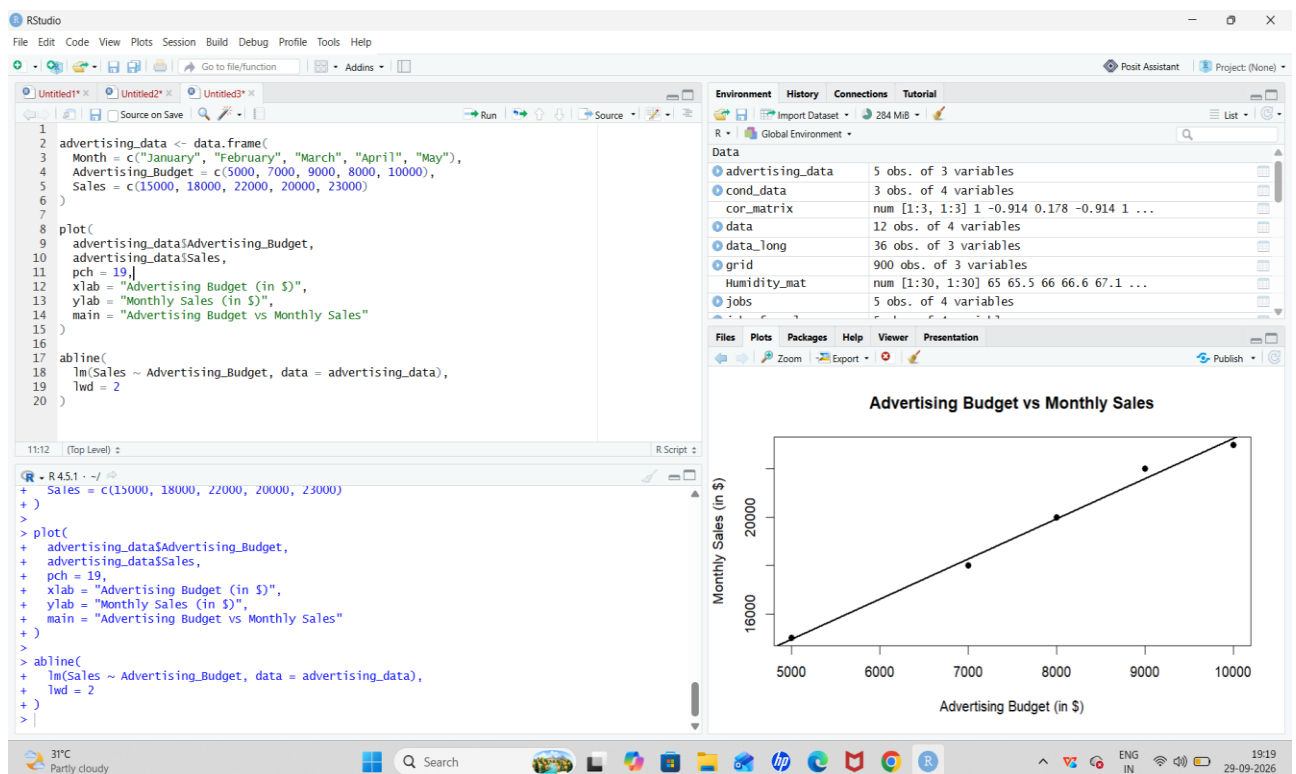


3. Develop a scatter plot to explore the relationship between advertising budget and monthly sales. Explain the insights drawn from the scatter plot.

```
advertising_data <- data.frame(
  Month = c("January", "February", "March", "April", "May"),
  Advertising_Budget = c(5000, 7000, 9000, 8000, 10000),
  Sales = c(15000, 18000, 22000, 20000, 23000)
)

plot(
  advertising_data$Advertising_Budget,
  advertising_data$Sales,
  pch = 19,
  xlab = "Advertising Budget (in $)",
  ylab = "Monthly Sales (in $)",
  main = "Advertising Budget vs Monthly Sales"
)

abline(
  lm(Sales ~ Advertising_Budget, data = advertising_data),
  lwd = 2
)
```



4. Build an interactive dashboard combining the line chart and bar chart to allow users explore sales data interactively.

```
library(shiny)
library(plotly)

sales_data <- data.frame(
  Month = c("January", "February", "March", "April", "May"),
  Sales = c(15000, 18000, 22000, 20000, 23000)
)

product_data <- data.frame(
  Product = c("Laptop", "Mobile", "Tablet", "Headphones", "Smartwatch"),
  Sales = c(45000, 38000, 30000, 22000, 18000)
)

ui <- fluidPage(

  titlePanel("Monthly Sales Dashboard"),

  sidebarLayout(

    sidebarPanel(
      selectInput(
        "month",
        "Select Month:",
        choices = sales_data$Month,
        selected = "January"
      )
    ),

    mainPanel(

      h3("Monthly Sales Trend"),
      plotlyOutput("lineChart"),

      h3("Top-Selling Products"),
      plotlyOutput("barChart")

    )
  )
)

server <- function(input, output) {

  output$lineChart <- renderPlotly({

    p <- plot_ly(
      sales_data,
      x = ~Month,
      y = ~Sales,
      type = "scatter",
      mode = "lines+markers"
    ) %>%

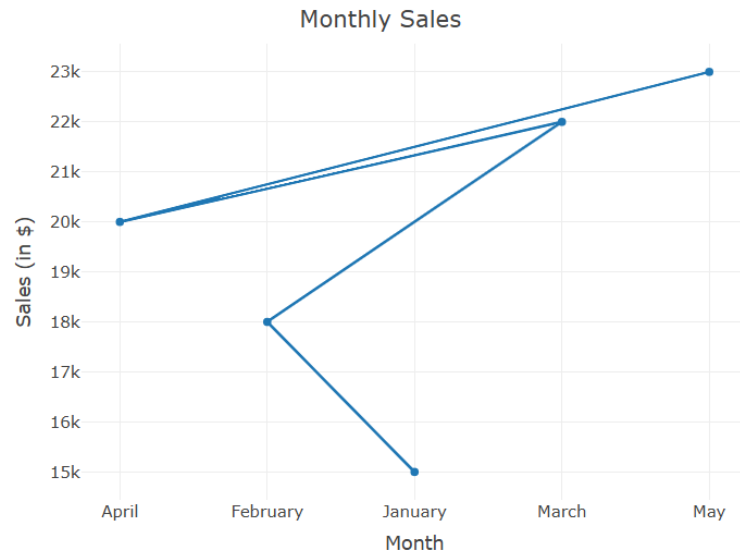
    layout(
      title = "Monthly Sales".
```

Monthly Sales Dashboard

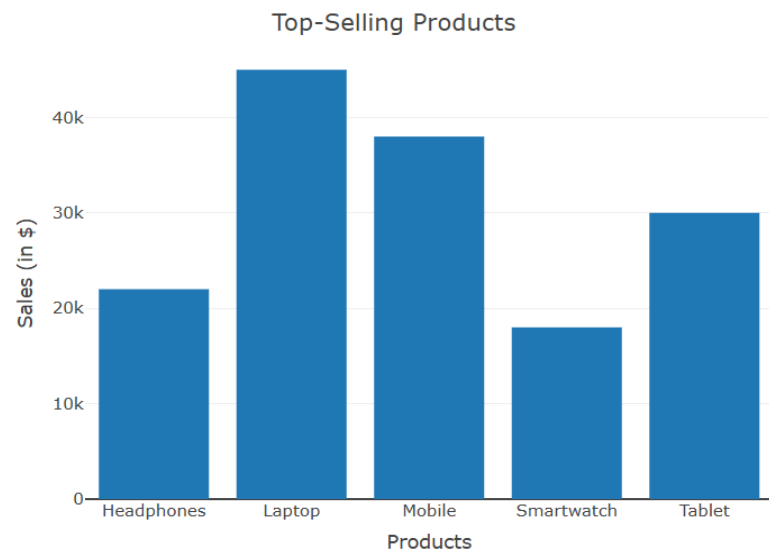
Select Month:

January

Monthly Sales Trend



Top-Selling Products



Procedure

1. Create the monthly sales dataset in R with month and sales values.
2. Use the plot() function to create a **line chart** showing monthly sales.
3. Create product-wise sales data and use the barplot() function to display the **top-selling products**.
4. Add advertising budget and sales data and create a **scatter plot** to analyze their relationship.
5. Use the abline() function to add a trend line to the scatter plot.
6. Install and load the **Shiny** and **Plotly** packages.
7. Create an interactive dashboard containing the monthly sales line chart and product sales bar chart.
8. Run the dashboard in RStudio and interact with the charts.

Result

The monthly sales data was successfully analyzed using R. The **line chart** displayed the sales trend from January to May, showing an overall increase in sales. The **bar chart** compared the sales of different products and identified the top-selling products. The **scatter plot** showed a positive relationship between advertising budget and monthly sales in the sample data. Finally, an **interactive dashboard** was developed using Shiny and Plotly to explore the sales information interactively.